

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10 – SUB-STRAND 2.2 REFLECTION: FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Driving question: Why does a mirror show a perfect copy of the world, and where exactly does that copy live? Use everything you have learned from Lesson 1 to Lesson 8 to answer the questions below. Write your answers in the space under each question. Refer to the bedroom mirror, the matatu rear-view mirror and the letter 'A' beside a plane mirror. Use diagrams where they help, and use mathematical words such as mirror line, object distance, image distance, perpendicular, congruent and coordinates. Show your working when you calculate or use coordinates.

Part 1: The puzzle and the mirror line

Aisha stands in front of her full-length bedroom mirror in Nairobi. She raises one hand and her reflection raises the opposite hand. The Kanga skirt pattern and the buttons on her shirt all look the same. In class, your teacher placed a bold letter 'A' beside a plane mirror and the image looked flipped left to right.

(a) What is a line of symmetry, and how is it connected to a mirror line?

(b) From your Symmetry Safari, how many lines of symmetry did a circle, a square, a rectangle, an equilateral triangle and a scalene triangle have?

(c) Explain how the fold test and the mirror test both give the same lines of symmetry.

Write your answer here:

Part 2: The three properties of reflection

In Lesson 2 you measured the position of images in a plane mirror and constructed image points with a ruler and set square.

(a) State the three properties of reflection on a plane surface.

(b) Otieno stands 1.2 m in front of a plane mirror. How far is he from his image? Show your working.

(c) Use all three properties to explain why Aisha's bedroom-mirror image looks the same size and shape as her, and appears directly across the mirror from her, even though it is laterally reversed.

Write your answer here:

Part 3: Reflection on the Cartesian plane

On the Cartesian plane, the y-axis ($x = 0$), the x-axis ($y = 0$) and the line $y = x$ can each act as a mirror line.

(a) Write the rule for reflecting a point (x, y) in each of these three lines.

(b) Point A(2, 5) is reflected in each of the three mirror lines. Write the coordinates of each image.

(c) Aisha stands at $x = 3$ in front of a mirror wall along the y-axis. Where is her image, and how does this explain why the image raises the opposite hand?

Write your answer here:

Part 4: Finding a hidden mirror line

In Lesson 6 you found the mirror line when the mirror had been removed.

- (a) A point P and its image P' are given on plain paper with no mirror shown. Describe how you construct the mirror line with a ruler and compass.
- (b) Explain why the mirror line must pass through the midpoint of PP' AND be perpendicular to PP'. What would go wrong with any other line through the midpoint?
- (c) On the Cartesian plane, how can you find the mirror line for an object point and its image by using algebra? Include what happens when the segment joining them is horizontal or vertical.

Write your answer here:

Part 5: Answering the driving question

Why does a mirror show a perfect copy of the world, and where exactly does that copy live?

Write a full explanation of about 6 to 8 sentences. Use the bedroom mirror, the matatu rear-view mirror and the letter 'A'. Include the unifying idea that connects the three properties of reflection. Give one real-life Kenyan application from Lesson 7 that works for the same reason.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Understanding of lines of symmetry and mirror lines	Defines a line of symmetry accurately, links it to the mirror line, gives all correct counts for the shapes and explains why the fold and mirror tests agree.	Defines a line of symmetry and gives most counts correctly, with a partial link to the mirror line.	Gives an incomplete definition or several incorrect counts, with no link between folding and mirroring.
Properties of reflection and calculations	States all three properties precisely and correctly calculates 2.4 m, using the properties to explain the phenomenon fully.	States the three properties and calculates the distance correctly, with a partial explanation of the phenomenon.	Names fewer than three properties or makes errors in the distance calculation.
Cartesian reflection rules	Gives all three rules correctly, finds all three images correctly and explains the lateral flip through the sign change.	Gives the rules and most images correctly, with a limited explanation of the flip.	Confuses the rules or gives incorrect image coordinates.
Finding the mirror line and reasoning	Describes the compass construction accurately, justifies the perpendicular bisector using equal distances and perpendicularity (showing why a non-perpendicular line fails), and gives the full algebraic method: midpoint, negative reciprocal gradient, $y - y_1 = m(x - x_1)$, and the horizontal and vertical special cases.	Describes the construction and gives a partial justification or most steps of the algebraic method.	Gives an unclear construction and no justification, or omits the perpendicular gradient.
Final explanation using Kenyan contexts	Gives a coherent, accurate explanation that links the perpendicular bisector to all three properties, the lateral reversal, the matatu mirror and a Kenyan application.	Gives an accurate explanation that covers most ideas with some contexts.	Gives an explanation that is incomplete or has misconceptions, with few links to the phenomenon.

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